The Cramér-Rao Lower Bound

Let $X_1, X_2, \ldots, X_n$ be a random sample from some distribution with pdf $f(x;\theta)$.

Consider estimating some $\tau(\theta)$.

Suppose that $\tau(\theta)$ is any unbiased estimator of $\tau(\theta)$.

Then $$\mathrm{Var}[\hat{\tau}(\theta)] \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)}$$ the CRLB

The Cramér-Rao Lower Bound

$$\mathrm{Var}[\hat{\tau}(\theta)] \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)}$$

Fisher information:

$$I_n(\theta) := \mathrm{E}\left[\left(\frac{\partial}{\partial\theta}\ln f(\vec{X};\theta)\right)^2\right]$$

Notation:

$$\mathrm{Var}[\widehat{\tau(\theta)}] \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)}$$

The estimator: $\tau(\theta) = T = t(X)$

so

$$\mathrm{Var}[T] \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)}$$

The proof of the Cramér-Rao Lower Bound depends on the Cauchy-Schwartz inequality:

$$\left(\int g(x)\, h(x)\, dx \right)^2 \leq \left(\int g^2(x)\, dx \right) \left(\int h^2(x)\, dx \right)$$

Proof: Consider

$$\int\int \big(g(x)h(y) - g(y)h(x) \big)^2 \, dx\, dy \geq 0$$

Square the left-hand side out and run the integrals through.

Get

$$0 \leq \int\int g^2(x)h^2(y)\,dx\,dy$$

$$-2\int\int g(x)h(y)g(y)h(x)\,dx\,dy$$

$$+\int\int g^2(y)h^2(x)\,dx\,dy$$

these are the same

Get

$$0 \le 2\int\int g^2(x)h^2(y)\,dx\,dy$$

$$-2\boxed{\int\int g(x)h(y)g(y)h(x)\,dx\,dy}$$

$$\int\int g(x)h(y)g(y)h(x)\,dx\,dy$$

$$= \int g(y)h(y)\left(\int\int g(x)h(x)\,dx\,dy\right) dy$$

$$= \left(\int\int g(x)h(x)\,dx\,dy\right)\left(\int\int g(y)h(y)\,dy\right)$$

$$\boxed{= \left(\int g(x)h(x)\,dx\right)^2}$$

Get

$$0 \le 2\int\int g^2(x)h^2(y)\,dx\,dy$$

$$-2\left(\int g(x)h(x)\,dx\right)^2$$

$$\left(\int g(x)h(x)\,dx\right)^2 \le \int\int g^2(x)h^2(y)\,dx\,dy$$

$$\le \left(\int g^2(x)\,dx\right)\left(\int h^2(y)\,dy\right)$$

$$\le \left(\int g^2(x)\,dx\right)\left(\int h^2(x)\,dx\right)$$

$$\left(\int g(x)\,h(x)\,dx\right)^2 \le \left(\int g^2(x)\,dx\right)\left(\int h^2(x)\,dx\right)$$

- This holds with sums. (discrete version)
- This holds with dx replaced by f(x)dx where f is a pdf:

$$(E[g(X)h(X)])^2 \le E[g^2(X)]\,E[h^2(X)]$$

$$\mathrm{Var}[\widehat{\tau(\theta)}] \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)}$$

$$I_n(\theta) := \mathrm{E}\left[\left(\frac{\partial}{\partial\theta}\ln f(\vec{X};\theta)\right)^2\right]$$

Will be done if we can show that:

$$\tau'(\theta) = \underbrace{\left(T - \tau(\theta)\right)}_{g(\vec{X})}\underbrace{\left(\frac{\partial}{\partial\theta}\ln f(\vec{X};\theta)\right)}_{h(\vec{X})}$$

where $T=t(\vec{X})$.

$$\tau'(\theta) = \left(\mathsf{T} - \tau(\theta)\right)\left(\frac{\partial}{\partial\theta}\ln \mathrm{f}(\vec{\mathsf{X}};\theta)\right)$$

$$\Downarrow$$

$$\mathsf{E}[\tau'(\theta)] = \mathsf{E}\left[\left(\mathsf{T} - \tau(\theta)\right)\left(\frac{\partial}{\partial\theta}\ln \mathrm{f}(\vec{\mathsf{X}};\theta)\right)\right]$$

$$\| \quad \| \quad \|$$

$$\tau'(\theta)$$

$$[\tau'(\theta)]^2 = \left(\mathsf{E}\left[\left(\mathsf{T} - \tau(\theta)\right)\left(\frac{\partial}{\partial\theta}\ln \mathrm{f}(\vec{\mathsf{X}};\theta)\right)\right]\right)^2$$

$$[\tau'(\theta)]^2 = \left(\mathsf{E}\left[\left(\mathsf{T} - \tau(\theta)\right) \left(\frac{\partial}{\partial\theta} \ln \mathsf{f}(\vec{\mathsf{X}}; \theta) \right) \right] \right)^2$$

$$\leq \mathsf{E}\left[\left(\mathsf{T} - \tau(\theta)\right)^2 \right] \cdot \mathsf{E}\left[\left(\frac{\partial}{\partial\theta} \ln \mathsf{f}(\vec{\mathsf{X}}; \theta) \right)^2 \right]$$

$$= \mathsf{Var}[\mathsf{T}] \cdot \mathsf{I}_n(\theta)$$

$$\Rightarrow \mathsf{Var}[\mathsf{T}] \geq \frac{[\tau'(\theta)]^2}{\mathsf{I}_n(\theta)}$$

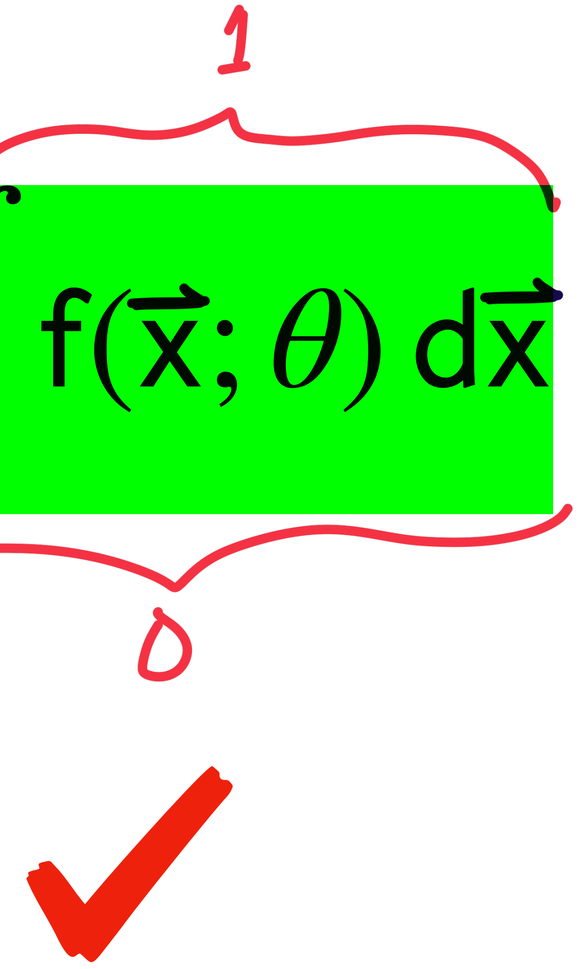
$$\tau'(\theta) = \left(\mathsf{T} - \tau(\theta)\right)\left(\frac{\partial}{\partial\theta}\ln \mathsf{f}(\vec{\mathsf{x}};\theta)\right)$$

$$\tau'(\theta) = \frac{\partial}{\partial\theta}\tau(\theta) = \frac{\partial}{\partial\theta}\mathsf{E}[\mathsf{T}]$$

$$= \frac{\partial}{\partial\theta}\int \mathsf{t}(\vec{\mathsf{x}})\,\mathsf{f}(\vec{\mathsf{x}};\theta)\,\mathrm{d}\vec{\mathsf{x}}$$

$$= \frac{\partial}{\partial\theta}\int \mathsf{t}(\vec{\mathsf{x}})\,\mathsf{f}(\vec{\mathsf{x}};\theta)\,\mathrm{d}\vec{\mathsf{x}} - \tau(\theta)\frac{\mathrm{d}}{\mathrm{d}\theta}\int \mathsf{f}(\vec{\mathsf{x}};\theta)\,\mathrm{d}\vec{\mathsf{x}}$$

$$= \int (\mathsf{t}(\vec{\mathsf{x}}) - \tau(\theta))\frac{\partial}{\partial\theta}\mathsf{f}(\vec{\mathsf{x}};\theta)\,\mathrm{d}\vec{\mathsf{x}}$$



$$\tau'(\theta) = \int (\mathrm{t}(\vec{\mathrm{x}}) - \tau(\theta)) \frac{\partial}{\partial\theta} \mathrm{f}(\vec{\mathrm{x}}; \theta)\, \mathrm{d}\vec{\mathrm{x}}$$

Want to see

- $$\mathrm{E}\left[\left(\mathrm{t}(\vec{\mathrm{X}}) - \tau(\theta)\right) \left(\frac{\partial}{\partial\theta} \ln \mathrm{f}(\vec{\mathrm{X}}; \theta) \right) \right]$$

- **Note that**

$$\frac{\partial}{\partial\theta} \mathrm{f}(\vec{\mathrm{x}}; \theta) = \frac{\partial}{\partial\theta} \ln \mathrm{f}(\vec{\mathrm{x}}; \theta)\, \mathrm{f}(\vec{\mathrm{x}}; \theta)$$

The CRLB is valid if

- $$\frac{\partial}{\partial\theta}\int f(\vec{x};\theta)\,dx = \int \frac{\partial}{\partial\theta} f(\vec{x};\theta)\,dx$$
- $\frac{\partial}{\partial\theta}\ln f(\vec{x};\theta)$ **exists**
- $$0 < E\left[\left(\frac{\partial}{\partial\theta}\ln f(\vec{X};\theta)\right)^2\right] < \infty$$

The CRLB is valid if

- $$\frac{\partial}{\partial\theta}\int f(\vec{x};\theta)\,dx = \int \frac{\partial}{\partial\theta} f(\vec{x};\theta)\,dx$$

This one doesn't hold whenever the parameter is in the indicator or support of the distribution.

Example:

The CRLB doesn't hold for the unif(0,θ) distribution!

$$\mathrm{Var}[\widehat{\tau(\theta)}] \geq \frac{[\tau'(\theta)]^2}{I_n(\theta)}$$

$$I_n(\theta) := E\left[\left(\frac{\partial}{\partial\theta}\ln f(\vec{X};\theta)\right)^2\right]$$

Example:

$$X_1, X_2, \ldots, X_n \overset{iid}{\sim} \text{Bernoulli}(p)$$

Find the Cramér-Rao lower bound of the variance of all unbiased estimators of p.

Here, θ=p and τ(p)=p.

The Fisher Information:

$$I_n(p) := E\left[\left(\frac{\partial}{\partial p} \ln f(\vec{X}; p)\right)^2\right]$$

pdf: $f(x; p) = p^x(1-p)^{1-x}\, I_{\{0,1\}}(x)$

joint pdf: $f(\vec{x}; p)$

$$= p^{\sum_{i=1}^n x_i}(1-p)^{n-\sum_{i=1}^n x_i} \prod_{i=1}^n I_{\{0,1\}}(x_i)$$

Take the log:

$$\ln f(\vec{x}; p)$$

$$= \left(\sum_{i=1}^{n} x_i \right) \ln p + \left(n - \sum_{i=1}^{n} x_i \right) \ln(1 - p)$$

Take the derivative:

$$\frac{\partial}{\partial p} \ln f(\vec{x}; p) = \frac{\sum_{i=1}^{n} x_i}{p} - \frac{n - \sum_{i=1}^{n} x_i}{1 - p}$$

Simplify:

$$\frac{\partial}{\partial p} \ln f(\vec{x}; p)$$

$$= \frac{(1-p)\sum_{i=1}^{n} x_i - p\left(n - \sum_{i=1}^{n} x_u\right)}{p(1-p)}$$

$$= \frac{\sum_{i=1}^{n} x_i - np}{p(1-p)}$$

Put the random variables in, square, and take the expectation.

Note that $Y = \sum_{i=1}^{n} X_i \sim \text{binomial}(n, p)$

$$I_n(p) = E\left[\left(\frac{\partial}{\partial p} \ln f(\vec{X}; p)\right)^2\right]$$

$$= E\left[\left(\frac{Y - np}{p(1-p)}\right)^2\right]$$

$$= \frac{1}{p^2(1-p)^2} \underbrace{E\left[(Y - np)^2\right]}_{\text{variance of binomial}}$$

Note that $Y = \sum_{i=1}^{n} X_i \sim \text{binomial}(n, p)$

$$I_n(p) = \frac{1}{p^2(1-p)^2} E\left[(Y - np)^2\right]$$

$$= \frac{1}{p^2(1-p)^2} \text{Var}[Y]$$

$$= \frac{np(1-p)}{p^2(1-p)^2} = \frac{n}{p(1-p)}$$

Example:

$$X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$$

$$\text{Var}[\hat{p}] = \frac{[\tau'(p)]^2}{I_n(p)}$$

$$= \frac{1^2}{n/[p(1-p)]}$$

$$= \frac{p(1-p)}{n}$$

Example:

$$X_1, X_2, \ldots, X_n \overset{iid}{\sim} \text{Bernoulli}(p)$$

mean: p

variance: $p(1-p)$

$$E[\overline{X}] = E[X_1] = p$$

$$\text{Var}[\overline{X}] = \frac{\text{Var}[X_1]}{n} = \frac{p(1-p)}{n}$$

So $\overline{X}$ is an unbiased estimator of p with the smallest possible variance! Woot!

This estimator for p for the Bernoulli distribution is actually a

Uniformly
Minimum
Variance
Unbiased
Estimator.

